



CALCULO INTEGRAL BMA02

semana 5

Método de integración:
De irracionales binomiales

$$I = \int x^m [a + bx^n]^p dx$$





Método de Chebyshev

$m, n, y p$ son racionales

$$I = \int x^m [a + bx^n]^p dx$$

caso 1: $\frac{m+1}{n} \in \mathbb{Z}$ $\Rightarrow x^n = t \Rightarrow nx^{n-1}dx = dt$

$$I = \frac{1}{n} \int t^{\frac{m+1}{n}-1} [a + bt]^p dt \Rightarrow a + bt = z^r \quad r \text{ es el denominador de } p$$

Ejercicio 1) calcule: $I = \int x^{-7} [65 - x^6]^{-\frac{1}{6}} dx$ $\Rightarrow m = -7, n = 6, p = -\frac{1}{6}$

$$\frac{m+1}{n} = -1 \in \mathbb{Z} \Rightarrow 65 - x^6 = z^6 \Rightarrow -6x^5 dx = 6z^5 dz \Rightarrow I = \int x^{-7} [z^6]^{-\frac{1}{6}} \left(-\frac{z^5}{x^5} \right) dz$$

$$I = - \int x^{-12} z^4 dz = - \int [65 - z^6]^2 z^4 dz = - \int (4225z^4 - 130z^{10} + z^{16}) dz$$

$$I = \frac{130}{11} \sqrt[6]{(65 - x^6)^{11}} - 845 \sqrt[6]{(65 - x^6)^5} - \frac{1}{17} \sqrt[6]{(65 - x^6)^{17}} + C$$



Método de Chebyshev

$m, n, y p$ son racionales

$$I = \int x^m [a + bx^n]^p dx$$

caso 2: $(\frac{m+1}{n} + p) \in \mathbb{Z}$ $\Rightarrow x^n = t \Rightarrow nx^{n-1}dx = dt$

$$I = \frac{1}{n} \int t^{\frac{m+1}{n}-1} [a + bt]^p dt \Rightarrow at^{-n} + b = z^r \quad r \text{ es el denominador de } p$$

Ejercicio 2) calcule: $I = \int t^2 [1 + t]^{\frac{1}{2}} dt \Rightarrow m = 2, n = 1, p = \frac{1}{2}$

$$1 + t = u^2 \Rightarrow dt = 2u du \Rightarrow I = \int [u^2 - 1]^2 [u^2]^{\frac{1}{2}} 2u du = 2 \int (u^6 - 2u^4 + u^2) du$$

$$I = 2 \left[\frac{u^7}{7} - \frac{2u^5}{5} + \frac{u^3}{3} \right] + C = \left[\frac{2(\sqrt{1+t})^7}{7} - \frac{4(\sqrt{1+t})^5}{5} + \frac{2(\sqrt{1+t})^3}{3} \right] + C$$



Método de Chebyshev

Ejercicio 3) calcule: $I = \int \frac{\sqrt[3]{1 + \sqrt[4]{x}}}{\sqrt{x}} dx$

$$I = \int x^{-\frac{1}{2}} [1 + x^{\frac{1}{4}}]^{\frac{1}{3}} dx \quad \Rightarrow \quad m = -\frac{1}{2}, n = \frac{1}{4}, p = \frac{1}{3} \quad \Rightarrow \quad \frac{m+1}{n} = 2$$

$$1 + x^{\frac{1}{4}} = u^3 \quad \Rightarrow \quad \frac{1}{4} x^{-\frac{3}{4}} dx = 3u^2 du \quad I = \int x^{-\frac{1}{2}} [u^3]^{\frac{1}{3}} 12u^2 x^{\frac{3}{4}} du$$

$$I = 12 \int x^{\frac{1}{4}} u^3 du = 12 \int [u^3 - 1] u^3 du = 12 \left(\frac{u^7}{7} - \frac{u^4}{4} \right) + C$$

$$I = 12 \left(\frac{[\sqrt[3]{1 + \sqrt[4]{x}}]^7}{7} - \frac{[\sqrt[3]{1 + \sqrt[4]{x}}]^4}{4} \right) + C$$



Método de Chebyshev

$$I = \int x^m [a + bx^n]^{\frac{p}{q}} dx$$

m, n, y p son racionales

caso 2: $\left(\frac{m+1}{n} + \frac{p}{q}\right) \in \mathbb{Z}$ $\Rightarrow u = \left[\frac{a + bx^n}{x^n}\right]^{\frac{1}{q}} \Rightarrow u^q = \left[\frac{a + bx^n}{x^n}\right]$

Ejercicio 4) calcule: $I = \int x^2 [3 + x^4]^{\frac{1}{4}} dx \Rightarrow u^4 = \left[\frac{3 + x^4}{x^4}\right] \Rightarrow 4u^3 du = [-12x^{-5}] dx$

$$I = \int x^2 [u^4 x^4]^{\frac{1}{4}} \left(-\frac{4}{12} x^5\right) u^3 du = -\frac{1}{3} \int x^8 u^4 du \quad u^4 = 1 + \frac{3}{x^4} \quad u^4 - 1 = \frac{3}{x^4}$$

$$I = -\frac{1}{3} \int \left[\frac{3}{u^4 - 1}\right]^2 u^4 du \Rightarrow I = -3 \int \left[\frac{1}{u^4 - 1}\right]^2 u^4 du$$



PROBLEMA 1

$$I_n = \int \frac{dx}{(1+x^2)^n}$$

Sea: $x = \tan \theta \quad \theta = \arctan x \rightarrow d\theta = \frac{dx}{1+x^2}$

$$dx = \sec^2 \theta d\theta$$

Reemplazando en la Integral

$$I_n = \int \frac{\sec^2 \theta d\theta}{(\sec^2 \theta)^n}$$


$$u = \sec^{-2n} \theta \rightarrow du = (-2n) \sec^{-2n} \theta (\tan \theta) d\theta$$

$$dv = \sec^2 \theta d\theta \rightarrow v = \tan \theta$$

Integración por partes

$$I_n = \sec^{-2n} \theta \tan \theta - \int \tan \theta (-2n \sec^{-2n} \theta \tan \theta d\theta)$$

$$I_n = \sec^{-2n} \theta \tan \theta + 2n \int \sec^{-2n} \theta \tan^2 \theta d\theta$$

$$I_n = \sec^{-2n} \theta \tan \theta + 2n \int \sec^{-2n} \theta (\sec^2 \theta - 1) d\theta$$

$$I_n = \sec^{-2n} \theta \tan \theta + 2n \left(\int \sec^{-2n} \theta \sec^2 \theta d\theta - \int \sec^{-2n} \theta d\theta \right)$$



$$I_n = \sec^{-2n} \theta \tan \theta + 2nI_n - 2n \int \cos^{2n} \theta \, d\theta$$

$$\text{Si } x = \tan \theta, \cos \theta = \frac{1}{\sqrt{x^2 + 1}}$$

Reemplazando la integral en funcion de x

$$I_n = \frac{x}{(1 + x^2)^n} + 2nI_n - 2n \frac{dx}{(1 + x^2)^n(1 + x^2)}$$

$$I_n = \frac{x}{(1 + x^2)^n} + 2nI_n - 2nI_{n+1}$$

$$I_n = \frac{1}{1 - 2n} \cdot \frac{x}{(1 + x^2)^n} - \frac{2n}{1 - 2n} I_{n+1}$$



PROBLEMA 2

$$I = \int \frac{dx}{(x^2 - 4x + 8)^{3/2}}$$

1er Método

$$I = \int \frac{dx}{((x - 2)^2 + 4)^{3/2}}$$

Sea:

$$x - 2 = 2 \tan \theta$$

$$dx = 2 \sec \theta^2 d\theta$$

$$I = \int \frac{2 \sec \theta^2 d\theta}{(4 \sec \theta^2)^{3/2}}$$

$$I = \frac{2}{8} \int \frac{\sec \theta^2 d\theta}{\sec \theta^3}$$

$$I = \frac{1}{4} \int \cos \theta d\theta$$

$$I = \frac{1}{4} \text{Sen} \theta + C$$

$$\theta = \arctan\left(\frac{x - 2}{2}\right)$$

$$I = \frac{1}{4} \text{Sen}\left(\arctan\left(\frac{x - 2}{2}\right)\right) + C$$



$$I = \int \frac{dx}{(x^2 - 4x + 8)^{3/2}}$$

2do Método

$$I = \int \frac{dx}{((x-2)^2 + 4)^{3/2}}$$

Sea: $x - 2 = 2\sinh\theta$
 $dx = 2\cosh\theta d\theta$

Recordar que: $\cosh^2\theta - \sinh^2\theta = 1$

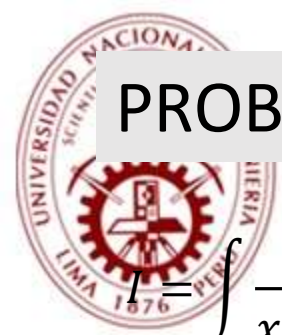
$$I = \int \frac{2\cosh\theta d\theta}{(4\cosh^2\theta)^{3/2}}$$

$$I = \frac{2}{8} \int \operatorname{sech}^2\theta d\theta$$

$$I = \frac{1}{4} \tanh\theta + C$$

$$\theta = \operatorname{arcsen} h x \left(\frac{x-2}{2} \right)$$

$$I = \frac{1}{4} \tanh(\operatorname{arcsen} h x \left(\frac{x-2}{2} \right)) + C$$



PROBLEMA 3

$$I = \int \frac{dx}{x^5 \sqrt{x^2 - 1}}$$

$$x = \frac{1}{t}, x = t^{-1}$$

$$dx = -1t^{-2}dt$$

$$I = \int \frac{-1t^{-2}dt}{t^{-5} \sqrt{t^{-2} - 1}}$$

$$I = - \int \frac{t^4 dt}{\sqrt{1 - t^2}} = - \left((At^3 + Bt^2 + Ct + D) \sqrt{1 - t^2} + \int \frac{\lambda}{\sqrt{1 - t^2}} \right)$$

$$A = -\frac{1}{4}$$

$$B = 0$$

$$C = -\frac{3}{8}$$

$$D = 0$$

$$\lambda = \frac{3}{8}$$





$$I = - \int \frac{t^4 dt}{\sqrt{1-t^2}} = - \left(\left(\frac{-1}{4} t^3 + \frac{-3}{8} t \right) \sqrt{1-t^2} + \int \frac{\frac{3}{8}}{\sqrt{1-t^2}} \right)$$

$$\int \frac{\frac{3}{8}}{\sqrt{1-t^2}} = (3/8)(\arcsen(t))$$

$$I = -\sqrt{1-t^2} \left(\frac{-1}{4} t^3 + \frac{-3}{8} t \right) - (3/8)(\arcsen(t))$$

$$t = x^{-1}$$

$$I = \sqrt{1 - \frac{1}{x^2}} \left(\frac{1}{4x^3} + \frac{3}{8x} \right) - \frac{3}{8} \arcsen \left(\frac{1}{x} \right) + c$$



PROBLEMA 5

$$I = \int \left(\frac{x+1}{x-1} \right)^{1/3} \frac{1}{x+1} dx = \int (x+1)^{-2/3} (x-1)^{-1/3} dx$$

$$x+1 = u, du = dx$$

$$I = \int (u)^{-2/3} (u-2)^{-1/3} du$$

$$u-2 = z^3 u$$

$$du = 2(1-z^3)^{-2} (3z^2) dz$$

$$I = \int (u)^{-2/3} (u-2)^{-1/3} du = \int 3z(1-z^3)^{-1} dz$$

$$I = 3 \int \frac{z dz}{1-z^3} = 3 \int \frac{z dz}{(1-z)(z^2+z+1)}$$

$$I = 3 \int \frac{1/3}{(1-z)} + 3 \int \frac{\left(\frac{1}{3}\right)z - 1/3}{(z^2+z+1)}$$



$$I_1 = 3 \int \frac{(1/3)dz}{(1-z)} = \int \frac{dz}{1-z}$$

$$I_1 = -\ln |1-z|$$

$$I_2 = 3 \int \frac{\left(\frac{1}{3}\right)z - 1/3}{(z^2 + z + 1)} = \frac{1}{2} \int \frac{2z - 2 + 1 - 1}{(z^2 + z + 1)}$$

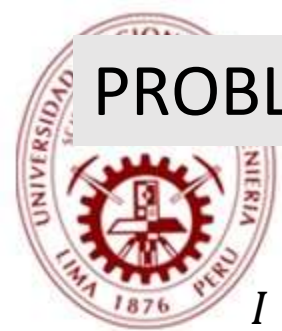
$$I_2 = \frac{1}{2} \int \frac{2z + 1}{(z^2 + z + 1)} + \frac{1}{2} \int \frac{-3}{(z^2 + z + 1)}$$

$$I_2 = \frac{1}{2} \ln |(z^2 + z + 1)| - \frac{3}{2} \int \frac{dz}{((z + 1/2)^2 + 3/4)}$$

$$I_2 = \frac{1}{2} \ln |(z^2 + z + 1)| - \sqrt{3} \arctan \left(\frac{2z + 1}{\sqrt{3}} \right)$$

$$I = I_1 + I_2 = \frac{1}{2} \ln |(z^2 + z + 1)| - \sqrt{3} \arctan \left(\frac{2z + 1}{\sqrt{3}} \right) - \ln |1 - z| \quad z = \sqrt[3]{\frac{x-1}{x+1}}$$

$$I = \frac{1}{2} \ln \left| \left(\sqrt[3]{\frac{x-1}{x+1}}^2 + \sqrt[3]{\frac{x-1}{x+1}} + 1 \right) \right| - \sqrt{3} \arctan \left(\frac{2 \sqrt[3]{\frac{x-1}{x+1}} + 1}{\sqrt{3}} \right) - \ln \left| 1 - \sqrt[3]{\frac{x-1}{x+1}} \right| + c$$



PROBLEMA 6

$$I = \int \sqrt{x^2 + x + 2 + \sqrt{2x^3 + 7x^2 + 4x + 3}} dx$$

Aplicando la formula de radicales dobles:

$$\sqrt{A \pm \sqrt{B}} = \sqrt{\frac{A+C}{2}} \pm \sqrt{\frac{A-C}{2}}$$

Donde: $C = \sqrt{A^2 - B}$

Sea: $A = x^2 + x + 2$ y $B = 2x^3 + 7x^2 + 4x + 3$

$$C = \sqrt{(x^2 + x + 2)^2 - (2x^3 + 7x^2 + 4x + 3)}$$

$$C = x^2 - 1$$

$$I = \int \left(\sqrt{\frac{x^2 + x + 2 + x^2 - 1}{2}} + \sqrt{\frac{x^2 + x + 2 - x^2 + 1}{2}} \right) dx$$

$$I = \int \frac{1}{\sqrt{2}} (\sqrt{2x^2 + x + 1} + \sqrt{x + 3}) dx$$

Sea: $I_1 = \int \sqrt{2x^2 + x + 1} dx$ y $I_2 = \int \sqrt{x + 3} dx$



$$I_1 = \int \sqrt{2x^2 + x - 1} \, dx$$

$$I_1 = \sqrt{2} \int \sqrt{\left(x + \frac{1}{4}\right)^2 + \frac{7}{16}} \, dx$$

$$\text{Sea: } x + \frac{1}{4} = \frac{\sqrt{7}}{4} \tan \theta \rightarrow dx = \frac{\sqrt{7}}{4} \sec^2 \theta \, d\theta$$

Reemplazando y operando:

$$I_1 = \frac{7\sqrt{2}}{16} \int \sec^3 \theta$$

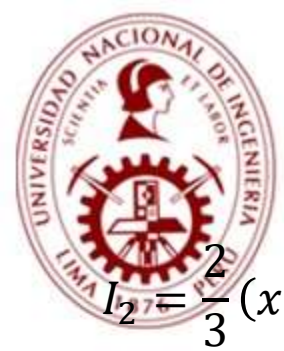
$$I_1 = \frac{7\sqrt{2}}{32} (\sec \theta \cdot \tan \theta + \ln |\tan \theta + \sec \theta|) + C$$

$$I_1 = \frac{7\sqrt{2}}{32} \left(\frac{\sqrt{(4x+1)^2 + 7} \cdot (4x+1)}{7} + \ln \left| \frac{\sqrt{(4x+1)^2 + 7} + 4x+1}{\sqrt{7}} \right| \right) + C$$

$$I_2 = \int \sqrt{x+3} \, dx$$

$$\text{Sea: } x+3 = u \rightarrow dx = du$$

$$I_2 = \int u^{1/2} \, du = \frac{2}{3} u^{3/2} + C$$



$$I_2 = \frac{2}{3}(x+3)^{3/2} + C$$

Entonces: $I = \frac{1}{\sqrt{2}}(I_1 + I_2)$

$$I = \frac{1}{\sqrt{2}} \left(\frac{7\sqrt{2}}{32} \left(\frac{\sqrt{(4x+1)^2 + 7} \cdot (4x+1)}{7} + \ln \left| \frac{\sqrt{(4x+1)^2 + 7} + 4x+1}{\sqrt{7}} \right| \right) + \frac{2}{3}(x+3)^{3/2} \right) + C$$